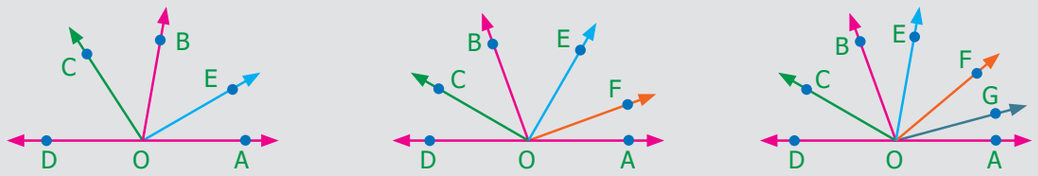
**Think**

What would happen to the angles if we add 3 or 4 or 5 rays on a line as given below?




We can learn one more result on linear pairs.
Observe the following Fig.5.9.
$AB$ is a straight line. $OC$ is a ray meeting $AB$ at $O$.
Here, $\angle AOC$ and $\angle BOC$ are linear pair.
Hence $\angle AOC + \angle BOC = 180°$
Also, OD is another ray meeting $AB$ at $O$.
Again $\angle AOD$ and $\angle BOD$ are linear pair.
Hence $\angle AOD + \angle BOD = 180°$
Now, $\angle AOC$, $\angle BOC$, $\angle AOD$ and $\angle BOD$ are the angles that are formed at the point $O$.
We can observe that $(\angle AOC + \angle BOC) + (\angle AOD + \angle BOD) = 180° + 180° = 360°$.
So, the sum of all angles at a point is 360°.

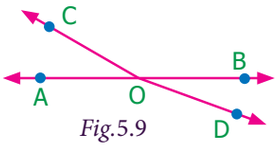
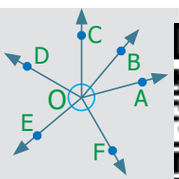


*Fig.5.9*

**Think**

Can you justify the following statement.
$\angle AOB + \angle BOC + \angle COD + \angle DOE + \angle EOF + \angle FOA = 360°$?




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**Example 5.6** From Fig.5.10, find the measure of $\angle ROS$.

***Solution***

We know that $\angle QOR + \angle ROS + \angle SOP = 180°$
$26° + \angle ROS + 32° = 180°$
$\angle ROS + 58° = 180°$
Subtracting 58° on both sides
We get, $\angle ROS = 180° - 58° = 122°$

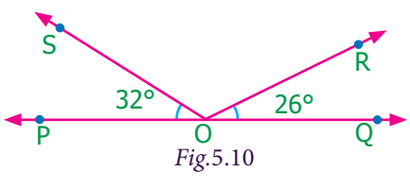


*Fig.5.10*

**Example 5.7** In Fig. 5.11, find the value of $x°$

***Solution***

$98° + 23° + 76° + x° = 360°$
$197° + x° = 360°$
$x° = 360° - 197° = 163°$

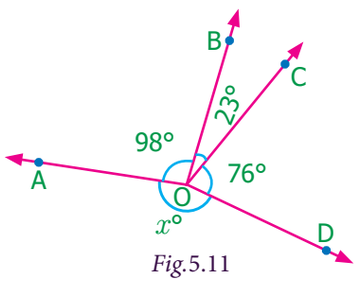


*Fig.5.11*

## 5.2.3 Vertically opposite angles

We have already studied about intersecting lines. Observe the Fig.5.12. There are two lines namely $l$ and $m$ which are intersecting at a point $O$ and forming four angles at that point of intersection. They are $\angle 1$, $\angle 2$, $\angle 3$ and $\angle 4$.

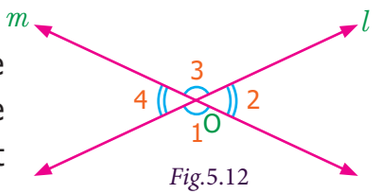


*Fig.5.12*



7th_Maths_T1_EM_Chp5.indd 90 10/12/2022 18:13:37